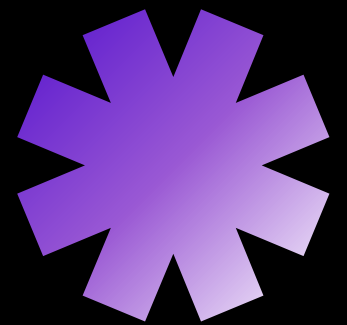
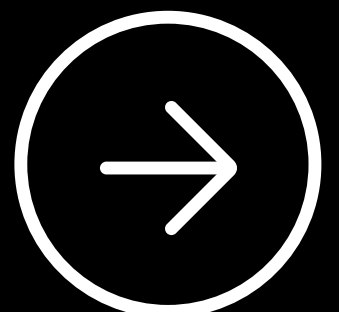


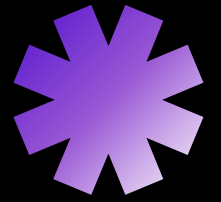


DEVOPS PRACTICE

DAY 16– DAY 21

- Shell scripting basics
- Loops and automation
- Functions and monitoring
- Backup and log rotation
- Log analyzer project
- Linux scripting cheat sheet



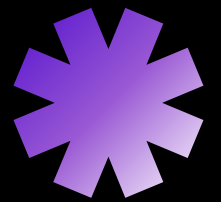


Shell Scripting Basics

16

- Created first Bash scripts
- Learned shebang (`#!/bin/bash`)
- Practiced variables and user input
- Used read command in scripts
- Practiced if-else conditions
- File existence checking using `-f`
- Combined scripts with `systemctl`

```
ubuntu@ip-172-31-3-219:~/devops$ ls
check_num.sh file.sh greet.sh hello.sh server_check.sh variable
ubuntu@ip-172-31-3-219:~/devops$ chmod +x hello.sh
ubuntu@ip-172-31-3-219:~/devops$ ./hello.sh
Hello ,DevOps!
ubuntu@ip-172-31-3-219:~/devops$ chmod +x variable.sh
ubuntu@ip-172-31-3-219:~/devops$ ./variable.sh
Hello, I am Asma and I am a DevOps Engineer
ubuntu@ip-172-31-3-219:~/devops$ chmod +x greet.sh
ubuntu@ip-172-31-3-219:~/devops$ ./greet.sh
Enter your name : Asma
Enter your favourite tool: Docker
Hello Asma, your favourite tool is Docker
ubuntu@ip-172-31-3-219:~/devops$ chmod +x check_num.sh
ubuntu@ip-172-31-3-219:~/devops$ ./check_num.sh
Enter a number: 1
Positive number
ubuntu@ip-172-31-3-219:~/devops$ ./check_num.sh
Enter a number: -1
Negative number
ubuntu@ip-172-31-3-219:~/devops$ ./check_num.sh
Enter a number: 0
Zero
ubuntu@ip-172-31-3-219:~/devops$ chmod +x file.sh
ubuntu@ip-172-31-3-219:~/devops$ ./file.sh
Enter filename: hello.sh
File exist
ubuntu@ip-172-31-3-219:~/devops$ ./file.sh
Enter filename: abc.txt
File does not exist
ubuntu@ip-172-31-3-219:~/devops$ chmod +x server_check.sh
ubuntu@ip-172-31-3-219:~/devops$ ./server_check.sh
Do you want to check the service status? (y/n): y
active
ubuntu@ip-172-31-3-219:~/devops$ ./server_check.sh
Do you want to check the service status? (y/n): n
Skipped
ubuntu@ip-172-31-3-219:~/devops$ ls -l
total 24
-rwxrwxrwx 1 ubuntu ubuntu 162 Jun  4 17:37 check_num.sh
-rwxrwxrwx 1 ubuntu ubuntu 125 Jun  4 17:40 file.sh
-rwxrwxrwx 1 ubuntu ubuntu 142 Jun  4 17:26 greet.sh
-rwxrwxrwx 1 ubuntu ubuntu  35 Jun  4 17:17 hello.sh
-rwxrwxrwx 1 ubuntu ubuntu 184 Jun  4 17:48 server_check.sh
-rwxrwxrwx 1 ubuntu ubuntu  90 Jun  4 17:20 variable.sh
ubuntu@ip-172-31-3-219:~/devops$
```



Loops, Arguments & Error Handling

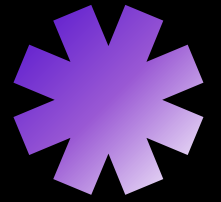
17

- Practiced for loop automation
- Used while loop for countdown
- Learned command-line arguments
- Used \$0, \$1, \$#, \$@
- Created package installer script
- Learned safe scripting practices
- Used set -e and error handling

```

Number : 6
Number : 7
Number : 8
Number : 9
Number : 10
ubuntu@ip-172-31-3-219:~/devops$ chmod +x countdown.sh
./countdown.sh
Enter a number: 5
Countdown: 5
Countdown: 4
Countdown: 3
Countdown: 2
Countdown: 1
Countdown: 0
Done!
ubuntu@ip-172-31-3-219:~/devops$ chmod +x greetAgain.sh
./greetAgain.sh
Usage: ./greet.sh <name>
ubuntu@ip-172-31-3-219:~/devops$ ./greetAgain.sh Asma
Hello, Asma!
ubuntu@ip-172-31-3-219:~/devops$ chmod +x args_demo.sh
./args_demo.sh apple mango banana
Script name: ./args_demo.sh
Total arguments: 3
All arguments: apple mango banana
ubuntu@ip-172-31-3-219:~/devops$ chmod +x install_packages.sh
./install_packages.sh
nginx is already installed
curl is already installed
wget is already installed
ubuntu@ip-172-31-3-219:~/devops$ chmod +x safe_script.sh

```



18- 19

Functions, Monitoring & Automation

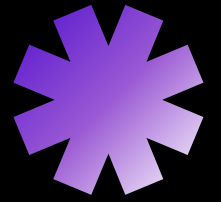
- Created reusable shell functions
- Learned strict mode concepts
- Used set -e, set -u, pipefail
- Checked disk and memory usage
- Created system information script
- Practiced local variables

```
ubuntu@ip-172-31-3-219:~/devops/script$ ls
disk_check.sh function.sh locl_demo.sh strict_demo.sh system_info.sh
ubuntu@ip-172-31-3-219:~/devops/script$ chmod +x function.sh
./function.sh
Hello, Asma!
Sum is : 15
ubuntu@ip-172-31-3-219:~/devops/script$ chmod +x disk_check.sh
./disk_check.sh
Disk Usage:
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        19G  2.6G   16G  15% /
Memory Usage:
              total        used        free      shared  buff/cache   available
Mem:          908Mi        349Mi        218Mi        2.7Mi        472Mi        559Mi
Swap:          0B           0B           0B
ubuntu@ip-172-31-3-219:~/devops/script$ chmod +x strict_demo.sh
./strict_demo.sh
Asma
ubuntu@ip-172-31-3-219:~/devops/script$ chmod +x locl_demo.sh
./locl_demo.sh
Inside function: Asma
Outside function:
ubuntu@ip-172-31-3-219:~/devops/script$ chmod +x system_info.sh
./system_info.sh
SYSTEM INFO
  Static hostname: ip-172-31-3-219
    Icon name: computer-vm
    Chassis: vm
Chassis Asset Tag: Amazon EC2
  Machine ID: ec2577353d8ee407689213af9795a96f
    Boot ID: bacd3c7f4d004f5481dd898cedela579
  Virtualization: amazon
Operating System: Ubuntu 26.04 LTS
    Kernel: Linux 7.0.0-1004-aws
  Architecture: x86_64
  Hardware Vendor: Amazon EC2
  Hardware Model: t3.micro
Firmware Version: 1.0
  Firmware Date: Mon 2017-10-16
  Firmware Age: 8y 7month 2w 5d

UPTIME
15:27:40 up 22:15,  1 user,  load average: 0.01, 0.04, 0.01

DISK USAGE
du: cannot read directory '/tmp/systemd-private-bacd3c7f4d004f5481dd898cedela579-systemd-hostname
d.service-YUeSjL': Permission denied
du: cannot read directory '/tmp/systemd-private-bacd3c7f4d004f5481dd898cedela579-chrony.service-G
09vHw': Permission denied
du: cannot read directory '/tmp/systemd-private-bacd3c7f4d004f5481dd898cedela579-ModemManager.ser
vice-9lnp4X': Permission denied
du: cannot read directory '/tmp/systemd-private-bacd3c7f4d004f5481dd898cedela579-systemd-logind.s
```

```
ubuntu@ip-172-31-3-219:~/devops/script$ touch myapps_logs/old.log
ubuntu@ip-172-31-3-219:~/devops/script$ ls myapps_logs
app.log error.log old.log
ubuntu@ip-172-31-3-219:~/devops/script$ vim log_rotate.sh
ubuntu@ip-172-31-3-219:~/devops/script$ chmod +x log_rotate.sh
ubuntu@ip-172-31-3-219:~/devops/script$ ./log_rotate.sh
./log_rotate.sh: line 5: $1: unbound variable
ubuntu@ip-172-31-3-219:~/devops/script$ vim log_rotate.sh
ubuntu@ip-172-31-3-219:~/devops/script$ ./log_rotate.sh
./log_rotate.sh: line 5: $1: unbound variable
ubuntu@ip-172-31-3-219:~/devops/script$ ./log_rotate.sh myapps_logs
./log_rotate.sh: line 7: LOG: unbound variable
ubuntu@ip-172-31-3-219:~/devops/script$ vim log_rotate.sh
ubuntu@ip-172-31-3-219:~/devops/script$ ./log_rotate.sh
./log_rotate.sh: line 5: $1: unbound variable
ubuntu@ip-172-31-3-219:~/devops/script$ ./log_rotate.sh myapps_logs
Checking log dir : myapps_logs
myapps_logs/old.log
myapps_logs/error.log
myapps_logs/app.log
Log check completed
ubuntu@ip-172-31-3-219:~/devops/script$ vim log_rotate.sh
ubuntu@ip-172-31-3-219:~/devops/script$ ./log_rotate.sh myapps_logs
Checking log directory: myapps_logs
Compressed files: 0
Deleted old compressed files: 0
Log rotation completed
ubuntu@ip-172-31-3-219:~/devops/script$ vim backup.sh
ubuntu@ip-172-31-3-219:~/devops/script$ chmod +x backup.sh
./backup.sh source_data backups
Backup completed
Backup file name: backup-2026-06-05.tar.gz
4.0K  backups/backup-2026-06-05.tar.gz
ubuntu@ip-172-31-3-219:~/devops/script$ ls backups
backup-2026-06-05.tar.gz
ubuntu@ip-172-31-3-219:~/devops/script$
```



Log Analyzer & Shell Scripting Revision

20
-21

- Analyzed Apache log files
- Counted error and critical logs
- Used grep, awk, sort, uniq
- Generated automated log reports
- Practiced real troubleshooting skills
- Reviewed daily Linux commands
- Practiced grep, awk, sed
- Learned cron and tar commands
- Revised scripting concepts
- Improved Linux troubleshooting confidence

```
asma@DESKTOP-07JRIII:~$ REPORT_FILE="log_report_$(date +%Y-%m-%d).txt"
asma@DESKTOP-07JRIII:~$ vim log_analyzer.sh
asma@DESKTOP-07JRIII:~$ ./log_analyzer.sh Apache_2k.log
Analyzing log file: Apache_2k.log

Total Error Count:
595

Critical Events:

Top 5 Errors:
368 mod_jk child workerEnv in error state 6
101 mod_jk child workerEnv in error state 7
44 mod_jk child workerEnv in error state 8
20 mod_jk child workerEnv in error state 9
12 mod_jk child init 1 -2

Report created: log_report_2026-06-05.txt
asma@DESKTOP-07JRIII:~$ ls
Apache_2k.log      ec2-volume-connect.pem  log_analyzer.sh      shell-scripting.pem
advanced-linux.pem ec2-volume.pem          log_report_2026-06-05.txt  snap
devops-practice.pem file.txt                script
asma@DESKTOP-07JRIII:~$ cat log_report_$(date +%Y-%m-%d).txt
Log Analysis Report
Date: Fri Jun 5 12:25:05 EDT 2026
Log File: Apache_2k.log
Total Lines: 1999
Total Error Count: 595

Top 5 Errors:
368 mod_jk child workerEnv in error state 6
101 mod_jk child workerEnv in error state 7
44 mod_jk child workerEnv in error state 8
20 mod_jk child workerEnv in error state 9
12 mod_jk child init 1 -2
```